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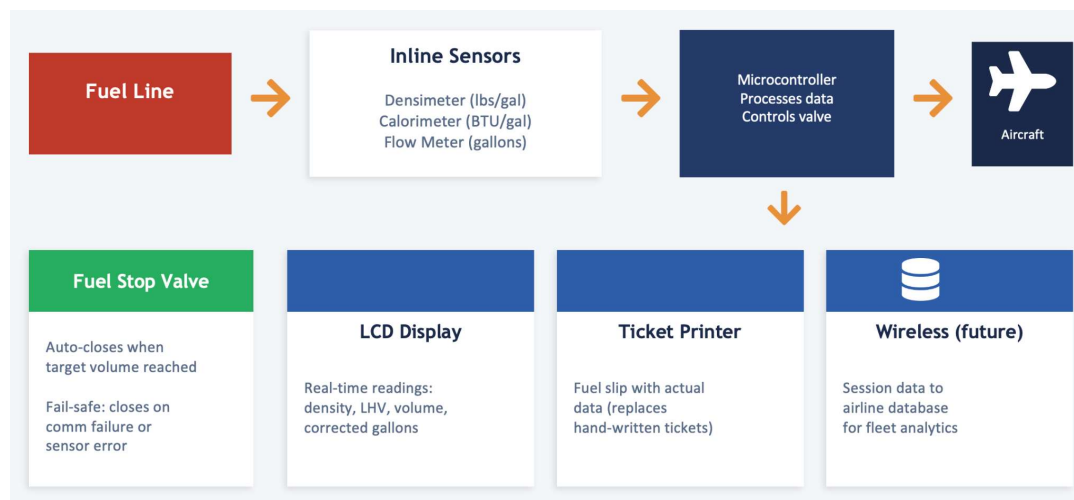
Repository: <https://github.com/nklloyd/flyprecise>

Alpha Test Plan

Fly Precise

Expected Behavior

1. **Use a flow meter with a calibrated screen for readings.** Use a flow meter that already has a display for readings to calibrate a flowmeter that has digital output.
2. **Source Code.** Source code for basic usage has given a baseline for how our sensors read data and output that data.
3. **Flow Chart.** A flow chart describing how the entire system will interact with a human user will be created in order to visually show operators how the system should respond to their manual inputs where needed.



Test Procedures

1. **Verify volumetric outputs match flow meter data.** Run the pump for a set measurement (e.g. 3 gallons) into a fixed container. Verify that the volumetric output of the flow meters is consistent with the output.

2. **Verify proximity measurements and specific gravity and density outputs are accurate.** Allow the hydrometer to settle in the density contraption and reference the output of the proximity sensor with a manual measurement via a ruler. Verify with a detached hydrometer and a scale measurement of a fixed volume to ensure the calculated density value from the attached system is accurate.
3. **Test integrated system with set mass values.** Input set mass values or their corresponding imperial weight (e.g. 10 Lbs., 15 Lbs., 20 Lbs.) and validate that the program shuts the pump off at the input mass. Document any margin of error to improve in future iterations.
4. **Test scaled down real-world fuel mass values.** Determine various aircraft, city-pairs, and total payload numbers for test flights. Based on estimated total fuel load, scale down the value by a factor of 100-1000 for testing practicality. Input the test value into the system and ensure the output is as expected.